Comparison of Artificial versus Spiking Neural Networks for Hardware-Constrained Edge Inference

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***Abstract*—This project compares artificial neural networks (ANNs) and spiking neural networks (SNNs) for hardware-constrained edge inference. The principal application is the classification of electrocardiogram arrhythmia, while handwritten digit recognition is kept as a reference task. Both models are evaluated using a PicoRV32 soft-core at 100 MHz, implemented on the programmable logic of a PYNQ-Z2 board. The comparison considers macro-F1 score, latency, throughput, computational cost, memory usage, and estimated power and energy. Upon conducting the comparison, it has been discovered that, depending on the application, both structures are feasible. Firstly, for dense inputs, ANN is superior to SNN, the MNIST analysis indicating SNN to be 5.5 times slower. Secondly, when applied to one beat at a time ECG inputs, SNN reaches the best macro-F1 score, but is 1.25 times slower. Lastly, for streamed ECG input, SNN is the suitable option, matching the macro-F1 of a GRU, but identifying heartbeats $\approx$ 100 times faster. Small custom coprocessors speed up the SNN three- to eightfold for less than 2% of the FPGA logic.**

***Index Terms*—artificial neural networks, edge AI, ECG classification, RISC-V, neuromorphic computing.**

I. Introduction

NEURAL networks (NN) are machine-learning models inspired by biological neurons. They are organized into layers and learn weights and biases that map inputs to outputs. NNs are widely used in pattern classification applications due to their trainable property, being present in computer vision, natural-language processing, and speech recognition [1], [2].

ECG arrhythmia detection is one of the key features in numerous wearable devices present on the current market, whose aim is to provide reliable monitoring of the wearer's heart beat and blood flow. In order to incorporate this function, power-efficient low-latency self-reliable processing units are needed to handle the identification and classification process of these patterns.

This project compares artificial neural networks (ANNs) and spiking neural networks (SNNs) for pattern recognition in ECG signals, with the goal of identifying selected arrhythmia classes, while also assessing which implementation is more suitable for the task. The design target is the inference unit of a battery-powered, wearable ECG monitor that classifies every heartbeat locally, within a fixed time budget and as energy efficient as possible. ANNs are the established choice for such classifiers, whereas SNNs process information as sparse events and are therefore expected to save time and energy on temporal signals such as ECG. Both network types are implemented on the same resource-constrained RISC-V soft core on a PYNQ-Z2 FPGA, the ANN serving as the baseline, and custom instructions are added to accelerate them.

The motivation behind this comparison is to determine the feasibility of different neural networks in edge AI computing. Producing resource-constrained devices capable of handling machine-learning workloads is one of the industry's main design challenges, as energy-efficiency is becoming more and more difficult to realize. While node sizing based reductions in energy consumption are approaching saturation, the demand for memory and faster data movement compromise the manufacturing capabilities for edge devices. Thus, replacements for canonical structures need to be developed. SNNs offer an event-driven computing model that neuromorphic hardware can exploit to reduce power consumption.

The ECG application is further described in Section II, including the envisioned monitoring system, the target metrics, and the semiconductor technology for a dedicated chip. Section III presents the experimental setup on the PYNQ-Z2, the data preparation, the network models and their training, using MNIST as a dense reference task next to the ECG. Finally, Section IV reports the measured quality, latency, and hardware cost, with Section V discussing them against the hypothesis and projecting the design to dedicated silicon.

As expressed before, the main attainment of this project is the comparison between ANN and SNN, on a restrictively structured processing unit. Moreover, this project is looking into the design of additional hardware support to further support SNN implementation:

> "For edge inference on a resource-constrained processor, is a spiking network (SNN) a better design choice than a classical ANN, and what hardware support is worth adding to make it advantageous?"

Thus, a hypothesis has been formulated. The expected outcome of this trial is that an ANN-based implementation is better suited for dense inputs (MNIST images), the SNN-based module better supports sparse, time-varying inputs (ECG signals), and a small hardware extension is expected to make the SNN cost-efficient. For the resource-constrained processing module implementation, a PicoRV32 soft RISC-V core is implemented in the FPGA fabric. This limited structure allows for the identification of the benchmark parameters using ANN, and also provides a starting point for the addition of supplementary, custom hardware extensions for SNN optimization.

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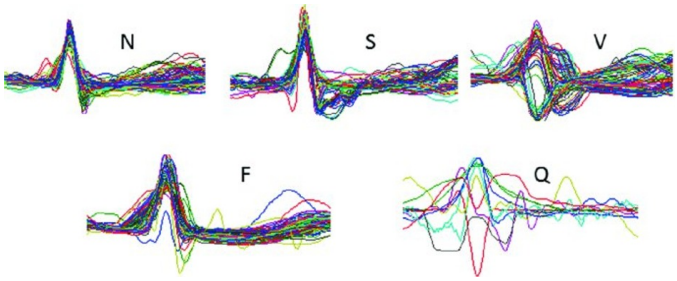


Fig. 1: ECG Signal Classes

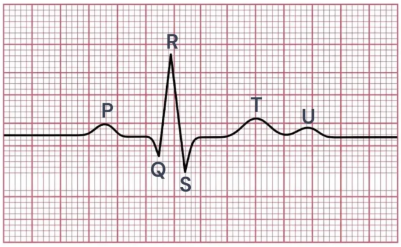


Fig. 2: ECG Signal Spike Classification

II. Application

A. Specific Application

The application scenario chosen for the basis of the comparison is the detection of different types of arrhythmia from an incoming dataset of ECG signals. The reasoning behind this choice lies in the topology of ECG signals – sparse, event-like - which makes them a favorable application for both continuous analysis (ANN) and event-driven analysis (SNN) [5]. Conducting ECG signal inference on two different computational units, one based on ANN, one based on SNN, provides sufficient insight into each system's performance metrics (latency, computational cost, accuracy) to then determine which architecture is best suited for ECG arrhythmia detection. For the scope of this project, three main types of arrhythmia will be detected: Normal (N), Supraventricular (S), Ventricular (V), as they have the highest occurrence rates (see Table I). In addition to the ECG scenario, the Modified National Institute of Standards and Technology (MNIST) dataset is used as a standardized reference task for comparing the two network types.

B. Underlying Theory

As mentioned before, ECG signals consist of patterns of sparse spikes, which, depending on the patient's condition may differ from the typical recording (See Figure 1). One sample spike is formed by 6 different peaks: P, Q, R, S, T, and U (see Figure 2) [7]. Depending on the duration and amplitude of these spikes, different cardiac conditions can be identified. Compared to the normal pattern, for supraventricular signals, the P spike is significantly reduced and the QRS complex is narrowed, due to the rapid electrical impulses generated above the ventricles. For ventricular signals, the P spike is almost absent, due to the wide band of the QRS complex, typically caused by a disruption in the heart's normal electrical pathway [8].

TABLE I: ECG Beat Classification Classes and Occurrences

Acronym	Class	Occurrence
N	Normal	90%
S	Supraventricular	3%
V	Ventricular	6%
F	Fusion	$< 1\%$
Q	Unclassifiable	$< 0.1\%$

C. State-of-the-Art

There are multiple examples across the industry of ECG monitoring units implemented using low-power ANN, particularly Convolutional Neural Networks (CNN) [9]. Z. Chen *et al.* (2023) reveals a low-power ECG processor ASIC based on a lightweight ANN, reaching a detection accuracy of 96.69% [10]. H. Mishra *et al.* (2026) presents a 14M-parameter CNN for binary AFib classification, trained on the HDXML dataset, achieving a calculated average F1-score of 77% [11]. K. Upreti *et al.* (2026) also describes a prototype of such a system, based on CNN, achieving through a benchmarking ECG dataset training a F1-score of 99.38% [12]. In the case of SNN-based SoC implementations of ECG processing, there are also numerous recent designs which can be mentioned. J. Mao *et al.* (2025) proposes a low-power SNN chip, the trained prototype on the MIT-BIH database achieving an accuracy of 91.4% [13]. O. Banjo *et al.* (2026) also presents a SNN implementation, for real-time ECG arrhythmia detection on edge hardware, using cascaded quantization and reaching a maximal F1 score of 91.8% [14].

Considering the current market, more and more recording dedicated devices are implementing neural networks in their processing units. Ambarella, Inc. is proposing a 4nm node, NN-based SoC for edge AI vision in their cameras [19]. Snapdragon W5+ also incorporates a dedicated ML accelerated module within its 22nm node coprocessor [20]. The Apple watch S10 includes a 4nm 4-core neural engine, being capable to record ECG signals and interpret the data for the users [21].

D. System Architecture

The proposed system is composed of signal acquisition, preprocessing and buffering, feature preparation, ANN or SNN inference, an output interface, and power management. The operating conditions below establish the processing, memory, and timing requirements for these subsystems.

The PYNQ-Z2 prototype evaluates inference using recorded data and prepared inputs. Physical acquisition and portable power management belong to the proposed device. MNIST is included as a separate reference workload.

1) Device and Acquisition Requirements: The proposed operating envelope assumes a minimum heartbeat separation of 250 ms, equivalent to 240 beats/min for regularly spaced arrivals. This exceeds the age-predicted maximal heart rate of adults, approximately $208 - 0.7 \times$ age and thus below 200 beats/min [32], so it bounds the beat rate the processing pipeline must sustain. Handling shorter separations would require a revised scheduling and queueing analysis.

One differential ECG channel is sampled at 500 samples/s. The Texas Instruments ADS1292R provides a reference acquisition solution: it integrates programmable-gain amplifiers,

24-bit delta-sigma ADCs, and an SPI interface [24]. Only one channel is required; the second channel and respiration function are outside the proposed application. Its ADC resolution is independent of the numerical precision used by the neural networks.

Each 24-bit sample is stored in a 32-bit word, giving a sample-storage rate of 2000 bytes/s. Actual SPI traffic also includes the device's status and framing information. Acquisition requirements are summarized in Table II. Patient-interface protection and physical acquisition are requirements of a future device, rather than implemented parts of the recorded-data prototype.

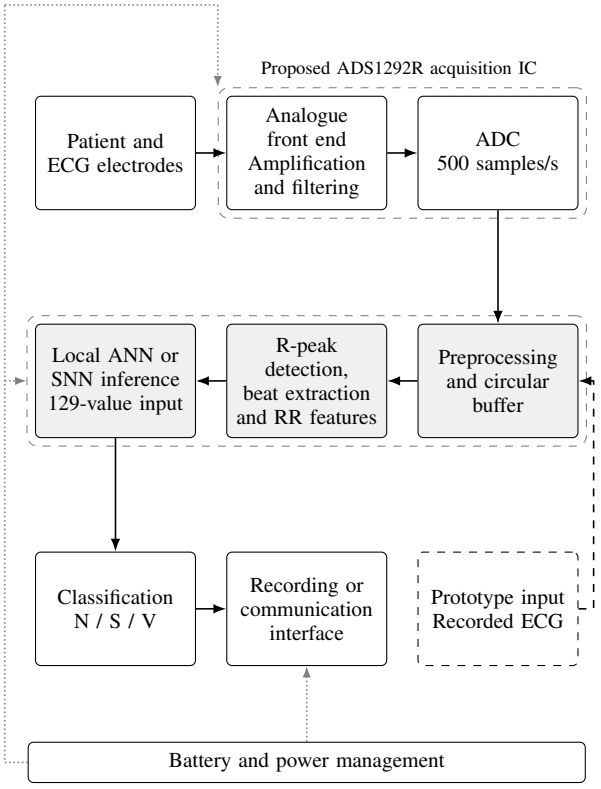


Fig. 3: Planned ECG monitoring system. Solid arrows represent signal flow, dotted arrows represent power distribution. Shaded blocks identify embedded processing. Recorded ECG replay (dashed) replaces physical acquisition in the prototype, the detector and buffering functions shown are requirements of the proposed device.

The proposed final device is a battery-powered ECG monitoring unit comprising electrodes, an acquisition front end, an inference processor, memory, and power-management circuitry. The acquisition front end, including its ADC, and the inference processor would be implemented as separate packaged ICs mounted on a common PCB, together with the required memory and power-management circuitry. Heartbeat classification would be performed locally, while a UART interface would allow classification results and timestamps to be transferred to a computer for logging and review. This physical arrangement is a proposed implementation concept. The PYNQ-Z2 experiments evaluate the inference subsystem and do not validate the complete acquisition, communication, or power-management system.

TABLE II: Proposed ECG Acquisition Requirements

Parameter	Proposed requirement
ECG channels	One differential channel
Sampling rate	500 samples/s
ADC output resolution	24 bits
Input-referred noise	≤ 10 μV peak-to-peak
Common-mode rejection	≥ 100 dB at 50/60 Hz
Acquisition IC power	< 1 mW in the reference config.
Digital interface	SPI

Noise is evaluated at gain 6 and 500 samples/s using the datasheet test configuration. Power refers to the acquisition IC with a 3 V analogue supply, a 1.8 V digital supply, and right-leg drive disabled. These are component-level requirements, not measured prototype performance or a whole-device power budget.

2) Proposed ECG Acquisition Technology: The Texas Instruments ADS1292R is selected as a reference component for the envisioned acquisition subsystem. It integrates two programmable-gain amplifiers, two 24-bit delta-sigma ADCs, and an SPI interface [24]. One differential channel is proposed; the second channel and respiration functionality are not required for this application. The ADC resolution does not imply 24-bit effective accuracy and does not determine the precision of network arithmetic.

Table II gives proposed acquisition requirements informed by the datasheet. At gain 6 and 500 samples/s, the datasheet reports approximately 8 μV peak-to-peak input-referred noise. It specifies a minimum common-mode rejection ratio of 105 dB at 50/60 Hz under its stated test conditions. At a 3 V analogue supply, with the right-leg-drive amplifier disabled, normal-mode chip power is 670 μW typical and 740 μW maximum [24]. These values exclude external processing and power-conversion losses.

Acquisition hardware, patient-interface protection, and the SPI acquisition path are not implemented by the recorded-data prototype. Their proposed specifications establish the context for an eventual physical monitoring device.

3) Signal Preparation and Buffering: The beat-window pathway uses 700 ms of ECG, extending from 250 ms before an R peak to 450 ms after it. At the proposed acquisition rate, this corresponds to 350 samples. To use the deployed model, the waveform must be converted to its training representation: 126 waveform values at an effective rate of 180 Hz, together with three RR-interval features, giving 129 inputs. The RR features are the previous interval, the next interval, and the running-mean interval. The waveform representation also subtracts a sample-wise median template formed from the previous 32 beats, as described in Section III-B.

The recorded-data experiments start from MIT-BIH recordings at 360 Hz. Their preprocessing is performed before inference, including filtering and down-sampling to 180 Hz. The proposed 500-sample/s acquisition rate therefore describes a future device and does not change the sampling rate or input representation of the reported experiments. A physical implementation would require validation of its filtering, resampling, normalization, template calculation, and quantization against the training pipeline.

A 512-sample circular acquisition buffer and a separate 350-sample processing buffer are proposed. At four bytes

per sample, these require 2048 and 1400 bytes, respectively, or 3448 bytes in total. The circular buffer holds 1.024 s of ECG and provides a nominal 324 ms storage margin beyond a complete 700 ms window. Completed windows must be extracted within this margin, with detection, scheduling, and copying delays accounted for.

R-peak timestamps identify pending windows. After waveform preparation, a compact representation of the beat can be retained while waiting for its next-RR feature, freeing the processing buffer for subsequent windows. This pending-beat storage and the 32-beat template history require separate allocations. For example, retaining 32 waveforms of 126 samples requires 4032 bytes at 8 bits/sample or 16128 bytes at 32 bits/sample, excluding workspace for the median calculation. The representation must preserve the preprocessing used by the model.

The zero-phase filtering used for recorded data is an offline operation. A live monitor would need a specified block-processing delay or a causal filter whose effect on classification is validated. Consequently, the prototype establishes inference performance on prepared inputs, rather than the end-to-end performance of a live acquisition pipeline.

4) Scheduling, Response Delay, and Power: The 250 ms minimum beat separation defines the proposed processing capacity requirement. Reserving up to 50 ms per beat for preparation, data movement, and acquisition-control overhead leaves an application budget of 200 ms for inference. This establishes the inference latency target of 200 ms per beat in Table III.

Inference throughput excludes acquisition and preprocessing time. An inference time of 200 ms corresponds to five prepared inputs per second, while including the reserved 50 ms overhead gives a complete pipeline capacity of four beats per second. Real-time scheduling must be checked against worst-case execution and acquisition-service times.

Classification response delay is different from processor execution time. The beat-window model requires 450 ms of post-R-peak signal and must also wait for the next heartbeat to obtain the next-RR feature. Ignoring filtering and detection delays, input availability is therefore no earlier than the larger of 450 ms and the next-RR interval. Queueing, preparation, inference, and output handling add further delay. The 200 ms inference target applies once the prepared input is available; it does not include the waiting time for waveform samples and RR features.

A battery and power-management subsystem would supply acquisition, processing, and the output interface. The ADS1292R's sub-milliwatt consumption is only the acquisition contribution.

E. Requirements and Initial Target Metrics

Initial target metrics establish a quantitative baseline for evaluating ANN and SNN implementations on PYNQ-Z2. The MNIST targets establish a reference benchmark for embedded classification, while the ECG targets are derived from the proposed monitoring architecture [Table III].

TABLE III: Initial Target Metrics

Metric	MNIST Target	ECG Target
F1 score	$\geq 95\%$	$\geq 65\%$
Inference latency	$\leq 40\,\mathrm{ms/image}$	$\leq 200\,\mathrm{ms/beat}$
Throughput	$\geq 25\,\mathrm{images/s}$	$\geq 5\,\mathrm{beats/s}$
Computational cost	101,632 MACs/image (ANN) 150,000 SOPs/image (SNN)	10,000 MACs/beat (ANN) 10,000 MACs + SOPs/beat (SNN)
Memory	$\leq 32\,\mathrm{KB}$	$\leq 32\,\mathrm{KB}$

1) F1 classification accuracy: For classification accuracy, the F1 score combines precision and recall using their harmonic mean. It ranges from 0 to 1, where 1 represents perfect classification performance. Macro-F1 calculates the F1 score for each target class and averages the values, giving equal importance to every class.

$$F_1 = 2 \cdot \frac{\mathrm{Precision} \cdot \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \tag{1}$$

$$F_{1,\mathrm{macro}} = \frac{1}{N} \sum_{i=1}^{N} F_{1,i} \tag{2}$$

As MNIST is a relatively simple standardized classification task, a 95% target F1 score provides a reasonable minimum for a small network constrained by the PYNQ-Z2 memory and computational power. For the ECG classification, a lower minimum threshold F1 score of 65% is chosen allowing for additional challenges of classifying different heartbeat morphologies (normal, supraventricular, ventricular).

2) Latency: Each MNIST input is a 28×28 grayscale image (784 pixel values) containing one handwritten digit from 0 to 9. A target of 40 ms/image means that each network must classify one prepared image within 40 milliseconds.

For the ECG, one input sample is a fixed-length segment of a continuous ECG signal, containing a single voltage measurement over time. The proposed architecture assumes a minimum heartbeat separation of 250 ms. From this, 50 ms is reserved for preprocessing, data flow, and acquisition-control overhead. Therefore, the remaining time establishes a 200 ms/beat inference latency target.

3) Throughput: Regarding the throughput, for both MNIST and ECG it's measured as the inverse of latency, corresponding to sequential processing. PicoRV32 soft-core processor executes one instruction at a time.

4) Computational cost: Computational cost is measured in multiply-accumulate operations (MACs) for ANNs and synaptic operations (SOPs) for SNNs. For MNIST, the 784–128–10 ANN architecture requires:

$$N_{\mathrm{MAC}} = 784 \times 128 + 128 \times 10 = 101,632 \text{ MACs/image.}$$

The MNIST SNN is assigned a preliminary target of 150,000 SOPs/image, accounting for rate encoding over 25 time-steps. For ECG, the ANN target is 10,000 MACs/beat, while the SNN has a combined budget of 10,000 MACs and SOPs per beat to account for direct encoding. These are proposed computational budgets and are the upper limits of the expected computational cost.

TABLE IV: Custom Instructions of the Coprocessors

Unit	Instruction	Operation
MAC	`mac.setx`	Load input value x
	`mac.acc4`	$a_{n+k} \mathrel{+}= x \cdot w_k$ for four packed weights
	`mac.get`	Read accumulator a_n
SNN	`snn.cfg`	Set decay β and threshold V_{th}
	`snn.acc4`	$I_{n+k} \pm= w_k$ for four packed weights
	`snn.step`	LIF update of up to 32 neurons
	`snn.get`	Read membrane potential or current

Clear and initialization instructions are omitted. All instructions take one cycle, except `snn.step` ($n + 4$ cycles for n neurons) and the clear instructions (n cycles).

5) Memory: The initial memory target is set to 32 KB to encourage compact implementation suitable for resource-constrained embedded systems, like this low-power/high-efficiency application. The target used for model evaluation is an inference-memory constraint, not the total memory capacity of the board or a demonstrated memory bound for the complete portable device.

F. Semiconductor Technology

The typical semiconductor technologies used for the implementation of neural networks include CMOS More than Moore (MtM) structures, featuring both analog and digital dies, to ensure high accuracy, while maintaining low power consumption. The technology node can vary between 4 and 180 nm nodes, since within this range cost-effectiveness and energy efficiency can be achieved [15]. Such nodes can be manufactured using Extreme Ultra Violet Lithography (EUV), for 4 nm to 7 nm nodes, or Deep Ultra Violet lithography (DUV), such as Argon Fluoride (ArF) lasers, for 7 nm to 90 nm nodes, or Krypton Fluoride (KrF) lasers, for 130 nm to 180 nm nodes. Standard material utilized for these systems is rigid Silicon wafers [16], featuring insulator substrates, silicon dioxide (SiO_2), and high-k dielectrics/ metal gates layers. There are also additional emerging implementations, such as the usage of 2D-materials (Molybdenum Disulfide (MoS_2), Hexagonal Boron Nitride ($h-BN$)) for atomic layer deposition, as they significantly lower the synaptic switching energy [17]. This will be further discussed in subsection V-B4.

Photonic neural-network accelerators are also being researched. They offer high speed for large matrix operations [18], but they are not yet manufactured commercially and are not suited to a small, low-power wearable device.

Thus, for the industry manufacturing process of the end-product of this project, the most suitable implementation technology is 65 nm node CMOS on Silicon. This scale allows for minimal area usage of the memory unit and greatly reduced dynamic power consumption (typically femtojoule range), while maintaining cost-efficiency and balanced mixed-signal matching.

III. Methodology

Both networks are evaluated using the same input sets, hardware platform, processor configuration, and measurement boundaries. The ANN establishes the baseline, after which the SNN and any SNN-specific hardware support are compared against that baseline.

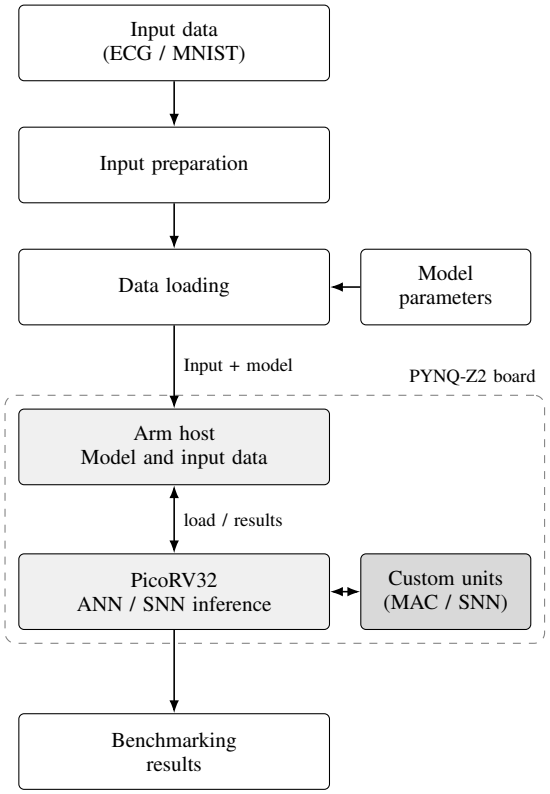


Fig. 4: Experimental prototype for ANN and SNN inference using prepared ECG inputs and MNIST images. The dashed enclosure identifies components on the PYNQ-Z2 board. The custom units execute the instructions of Table IV.

A. Experimental Setup

All experiments run on a PYNQ-Z2 development board, which is based on the Xilinx Zynq-7000 XC7Z020 System-on-Chip (SoC) [6]. The chip combines a dual-core Arm Cortex-A9 processor with programmable FPGA logic. A PicoRV32 RISC-V soft core, clocked at 100 MHz, is implemented on the FPGA and performs all neural-network inference. The Arm processor serves only as the host, loading the prepared inputs and the trained parameters into the memory of PicoRV32, starting inference, and collecting the classification outputs with the measured cycle count (Fig. 4). The deployed models use 8-bit integer weights and inputs, with the SNN additionally storing the membrane potential of every neuron. The embedded implementation is verified against an 8-bit reference model on the PC (Section III-E).

To accelerate inference, two coprocessors are attached to PicoRV32 through its coprocessor interface (PCPI) and are used through custom instructions (Table IV). The MAC unit multiplies one input value with four packed 8-bit weights and accumulates the products in a single instruction. It executes the dense layers of the ANN and the GRU, and the first layer of the direct-encoded SNN. The SNN unit adds the weights of a spiking input to the input currents of four neurons at once and performs the leaky integrate-and-fire update of Eq. (3) in a pipeline that processes one neuron per clock cycle. Both units skip zero-valued inputs, so only non-zero inputs and spikes

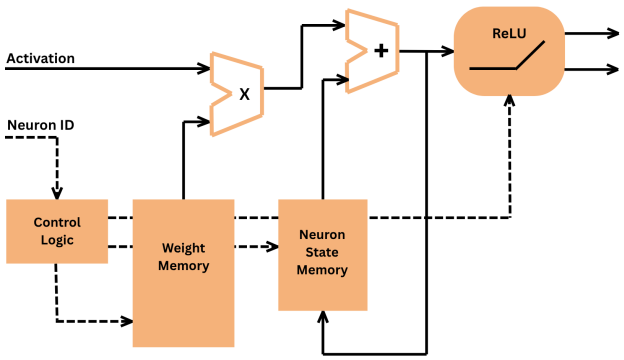


Fig. 5: Conceptual single-neuron datapath of the ANN: inputs are multiplied by their weights and accumulated (the state memory holds the partial sum), followed by the ReLU activation.

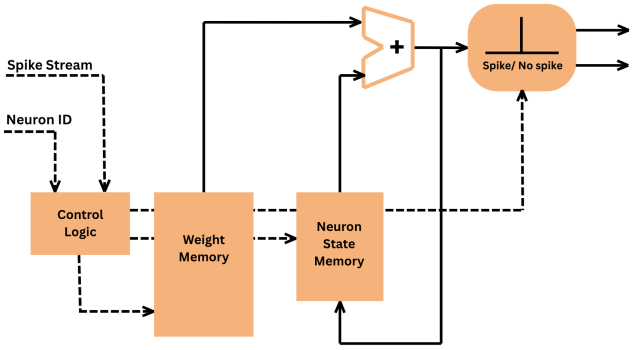


Fig. 6: Conceptual single-neuron datapath of the SNN: incoming spikes select weights that are added to the membrane potential, which is decayed by β and compared with the threshold (Eq. 3) to generate a spike.

cause work. Their FPGA cost is listed in Table VIII.

B. Data Preparation

The MIT-BIH recordings (lead MLII, 360 Hz) are initially band-pass filtered (0.5 to 40 Hz) with a zero-phase third-order Butterworth filter, before down-sampling by a factor of two to 180 Hz. Since the signal is already band-limited to 40 Hz, the reduced Nyquist frequency of 90 Hz loses no information while halving the input width. Each beat window spans 250 ms before and 450 ms after the annotated R peak, resulting in 126 samples per beat. To reduce patient-specific morphology, the network receives the difference between the input beat and a causal template, a sample-wise median of the previous 32 beats of the same recording. Three RR features (previous, next, and running-mean RR interval) are added, giving 129 inputs. With this input size, both ECG networks meet the memory and latency targets, so no further down-sampling was applied to the deployed models.

For the main ECG comparison, input preparation produces the 129-value beat representation. The additional recurrent-SNN and GRU comparison instead presents ECG samples sequentially and retains recurrent state; it does not use the same fixed-vector inference path. MNIST supplies prepared images through the same general loading and benchmarking framework.

C. Network Models

Artificial neural networks (ANNs) consist of layers of interconnected nodes that transform inputs using learned weights, biases, and activation functions. Their inference workload is commonly dominated by multiply-accumulate operations (see Fig. 5). ANNs are effective for static image recognition, but dense evaluation can perform unnecessary work when temporal inputs are sparse [3].

Spiking neural networks (SNNs) use state-responding neurons and discrete events called spikes (see Fig. 6). This event-driven representation can reduce synaptic activity when the input is sparse. However, any claimed latency or energy advantage must also account for input encoding, neuron-state updates, memory use, and the cost of executing synaptic operations [4].

In this project, the SNN uses the leaky integrate-and-fire (LIF) neuron model. At each timestep, the membrane potential decays, integrates the incoming input, and generates a spike when a defined threshold is exceeded:

$$\begin{aligned} u_t &= \beta v_{t-1} + I_t, \\ s_t &= H(u_t - V_{\text{th}}), \\ v_t &= u_t - s_t V_{\text{th}}, \end{aligned} \tag{3}$$

where v_t is the updated membrane potential, β is the decay factor, I_t is the input current, V_{th} is the firing threshold, and s_t is the generated spike. The function H represents the Heaviside step function.

For MNIST classification, the input images are converted into spike trains over multiple timesteps using rate encoding. For ECG classification, direct encoding is applied, feeding the quantized input values directly into the SNN without conversion into spike trains.

For streamed ECG, two recurrent networks with 64 hidden units receive the beat one sample at a time instead of as a complete window: a gated recurrent unit (GRU), a common recurrent ANN, and a recurrent SNN. The recurrent SNN uses delta encoding (Section III-D3), so it receives an input event only when the signal has changed by more than a threshold. At the end of each beat, an output layer classifies it from the network's internal state and the three RR features.

D. Model Design Studies

The encoding and down-sampling studies in this and following subsection were conducted on the PC with floating point weights, using validation macro-F1 over five seeds with 150 training epochs each. They ran in parallel with the deployment work and therefore use an exploratory configuration that differs from the final board model. The parameters of the deployed networks were fixed during the on-board measurements described in Section IV.

The ECG input must be represented digitally before it is supplied to the SNN. Two alternatives are compared.

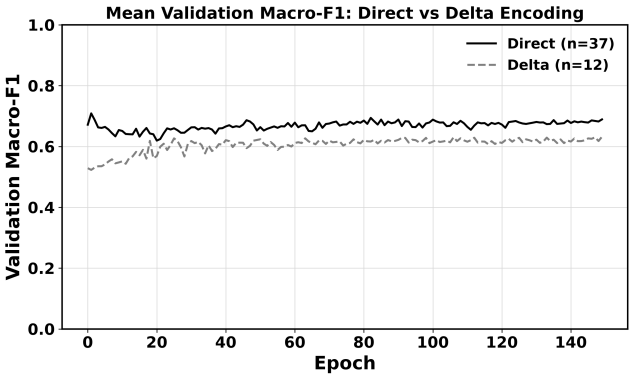


Fig. 7: Delta vs Direct encoding

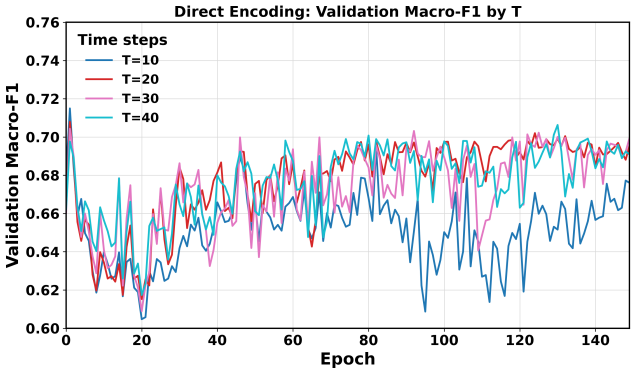


Fig. 8: Direct encoding with different T

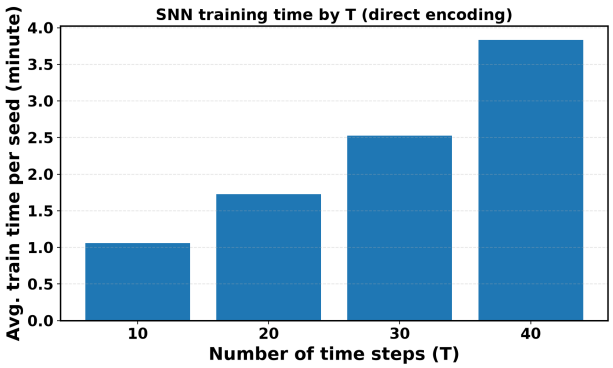


Fig. 9: Direct encoding T's with training times

1) Direct encoding: supplies the prepared input vector x (Sec. III-B) unchanged as an input at every timestep to the SNN, without applying an event threshold.

$$X(t) = x \mid t = 1, 2, ..., T, \tag{4}$$

T denotes the number of SNN simulation timesteps over which the complete input vector is repeatedly presented and is by itself an adjustable parameter of this encoding method.

2) Delta encoding: emits an event only when the change in signal magnitude exceeds a selected threshold θ. The value n denotes the index of a sample in the prepared ECG signal. Important difference between the delta and direct encoding is that in delta encoding the number of encoding timesteps is only dependent on the length of prepared ECG signal and not on an arbitrary variable T like in direct encoding.

$$\Delta x_n = x_n - x_{n-1}$$

$$s_n^+ = \begin{cases} 1 & \Delta x_n \geq \theta \\ 0 & \text{otherwise} \end{cases} \qquad s_n^- = \begin{cases} 1 & \Delta x_n \leq -\theta \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

The adjustable parameter here is the threshold θ, which allows us to determine how big of a change in signal we need for the spike to be emitted.

3) Choice of encoding: In order to determine which encoding will be the best choice for our use-case we set up an experimental setup to train the SNN over 150 epochs across different seeds while sweeping the values of T between 10 and 40 and the value of θ between 0.2 and 1.0. Fig. 7. shows us the average macro-F1 performance shown by the different encoding methods during the experiment. Every tested direct-encoding configuration had a higher mean macro-F1 than every tested delta-encoding configuration. The values of T greater than 40 will not be tested because of the increased duration in training time, which we noticed running smaller tests before this one.

4) Direct encoding specification: Since we already established that direct encoding performs better for our use-case with any setting tested, we may now analyze the different runs of the direct encoding to choose the best performing one. Fig. 8 shows us that the increase in value of T past the value of 20 does not come with any performance benefit. What is more, according to Fig. 9. we can clearly see that time needed to train the SNN scales linearly with the value of T. That is why amongst all tested values, we decided to choose the value $T = 20$ for our further optimization experiments. $T = 30$ and $T = 40$ show bigger performance variations between the epochs than $T = 20$ did and take longer to train per epoch. $T = 20$ is therefore used for the remaining studies in the subsection.

5) Down-sampling: In order to reduce input size of our signal and its computational cost a down-sampling factor DS can be chosen. Fig. 10. shows the value of validation macro-F1 for DS values swept between 2 and 12. This corresponds with 90 samples down to 15 samples of the ECG respectively. Original signal has 180 samples. We will test the performance across 150 epochs across 5 random seeds with $T = 20$. According to the results presented in Fig. 10., it can be clearly stated that with the down-sampling factor $DS = 6$ our SNN presents the best validation macro performance even taking into account the performance variation between the seeds. $DS = 6$ can thus be treated as our go-to operating point for the SNN, since we want to maximize its performance. A possible explanation to our observations is that the down-

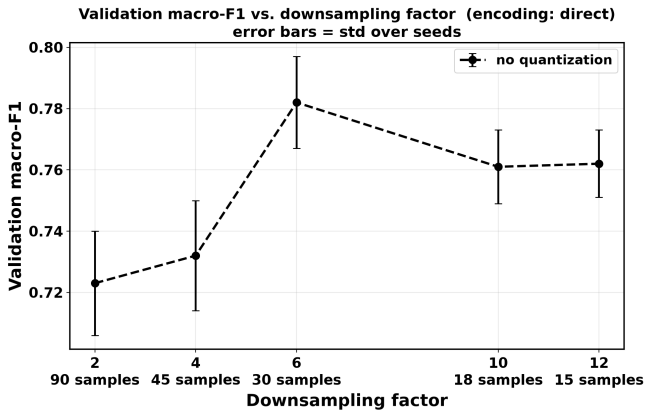


Fig. 10: Validation macro-F1 as a function of the down-sampling factor DS (mean over 5 seeds; error bars show the seed variation).

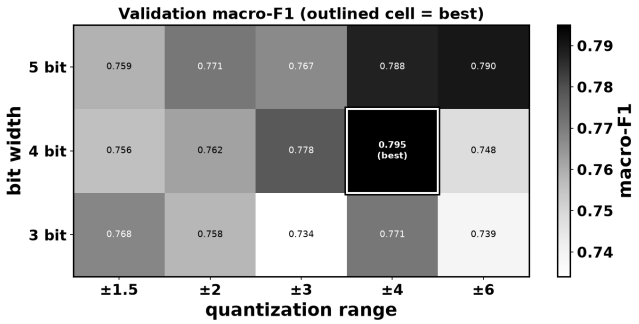


Fig. 11: Validation macro-F1 of the SNN for different quantization bit widths and quantization ranges.

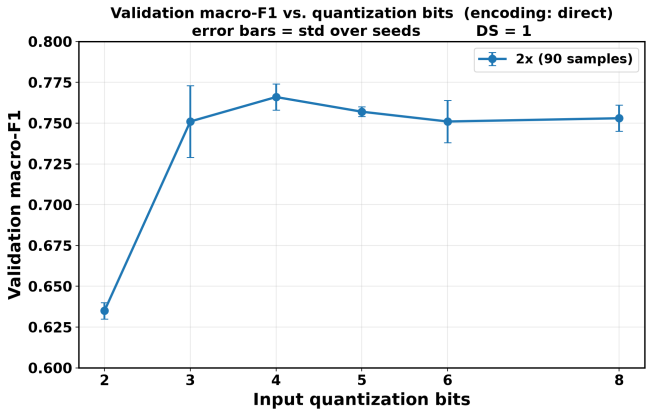


Fig. 12: Validation macro-F1 of the SNN for different quantization bit widths Q with $DS = 1$ down-sampling applied (mean over 5 seeds).

sampling discards minor signal variations which do not help the SNN analyze the signal but only introduce confusion to the model. $DS = 6$ is therefore the best setting in this study.

6) Quantization: Next, we will discuss introducing the quantization to our model. Since our model should run well on minimal hardware as listed in this paper, we need to make sure it utilizes only the resources it absolutely needs to. Hence, we will introduce input quantization in order to help save memory. The quantization will divide the signal-space in a number of 2^Q of quantization levels each spanning a range $R = [+x; -x]$ which will be denoted as $R = \pm x$ for simplicity. In this case the value Q represents the number of quantization bits. All ECG values are to be approximated to fit into those sections which will help us quantize the ECG vector values into single bit numbers. The experimental setup will remain as stated earlier with 150 epochs and 5 random seeds and T=20. The results of the experimentation can be seen on Fig. 11. After analyzing the results, we can clearly conclude that the best performing quantization setup across all of our seeds is when $Q = 4$ and $R = \pm 4$. As a reassurance test whether the results of quantization would overlap with down-sampling, we run another test setup with $R = \pm 4$ but with $DS = 1$ and the results are presented on Fig. 12. There, it can be clearly seen that Q=4 is still the best performing, but the overall performance is lower than with $DS = 6$. The best setting in this study is therefore $DS = 6$ $Q = 4$ and $R = \pm 4$.

In this study, down-sampling and input quantization improved the validation macro-F1 while reducing the input size. The deployed models of Section IV use $T = 10$, decimation by two and 8-bit quantization, since these studies ran in parallel with the deployment. Applying the settings found here is left for future work.

7) Training ANN and SNN: Both models are trained on the beat representation from Section III-B, and the SNN uses the direct encoding selected in Section III-D3.

The ECG training set is strongly imbalanced, with roughly 58 N beats for every S beat. Without any correction, the network can reach a low loss by predicting N almost all the time, which gives a high accuracy but detects hardly any arrhythmias. To counter this, we weight the square root of the inverse class frequency, which emphasizes the minority classes without making the training unstable.

Hyperparameters were selected by five-fold patient-level cross-validation on the training records using macro-F1, after which the final models were trained on all training records for a fixed 10 epochs. Macro-F1 is used instead of accuracy because a model that always predicts N already scores about 90% on the test set while detecting no arrhythmias at all. Such a model would only reach a macro-F1 of about 0.32. Macro-F1 averages the F1 score of the three classes with equal weight, so a model also has to do reasonably well on S and V to score high, which fits the goal of the classifier better.

The training and test patients do not overlap (the standard inter-patient split of de Chazal *et al.* [22]). This is a harder evaluation than the intra-patient splits used in many of the works in Section II-C, where beats from the same patient can end up in both the training and the test set. With the inter-patient split the model cannot rely on patient-specific waveform details, so it has to generalize to unseen patients. This is probably one of the reasons why our F1 scores are lower than the published numbers, although other differences also play a role, such as the metric (several of those works

TABLE V: PYNQ-Z2 Hardware Specifications

Component	Specification
Programmable Logic	13,300 logic slices
Block RAM	630 KB
DSP Slices	220
Clock Source	100 MHz

report accuracy) and the task itself.

E. Benchmarking on PYNQ-Z2

The ANN and SNN implementations are benchmarked on the PYNQ-Z2 board using the PicoRV32 RISC-V soft-core processor. To ensure a fair comparison, both models are evaluated using equivalent input datasets, processor clock frequencies, compiler settings, and measurement procedures. Classification performance is evaluated using the macro-F1 score, together with per-class precision and recall. Hardware performance is assessed through inference latency, throughput, computational cost, and memory usage.

Four implementation levels are considered. First, a straightforward C implementation provides a baseline without specific software optimizations. Second, an optimized C implementation improves execution efficiency through software-level modifications. Third, the optimized implementation is evaluated with an additional hardware coprocessor to investigate the benefits of hardware acceleration. Finally, the optimized muRISCV-NN library is used as an external reference for ANN inference, providing a comparison against an established software implementation.

To verify functional correctness, the classification outputs obtained on the PYNQ-Z2 are compared against the corresponding PC reference model for every test input. Identical outputs are required to confirm that the embedded implementation preserves the reference model's functionality.

Inference latency is measured in processor clock cycles, using a consistent measurement interval from the beginning of inference to the availability of the classification output. The execution time is calculated as:

$$t_{\text{inf}} = \frac{N_{\text{cycles}}}{f_{\text{clk}}}, \tag{6}$$

where N_{cycles} represents the measured processor clock cycles and f_{clk} is the processor operating frequency. Throughput is determined from the number of completed inferences per unit of time.

Power consumption is estimated using Vivado power reports rather than direct board-level measurements. Energy per inference is estimated from power and execution time, rather than measured experimentally.

IV. Results

Table VI summarizes the measured results of both ANN and SNN networks on each dataset, run on the PicoRV32 at 100 MHz. Compared with the initial targets of Table III, all networks meet the macro-F1 and throughput targets. With the coprocessor, all networks also meet the mean latency target, but the MNIST SNN exceeds it in the worst case. The ECG models meet the 32 KB memory target, while the MNIST models exceed it.

TABLE VI: Final ANN–SNN Comparison
(Board, PicoRV32 at 100 MHz, Mean over Seeds, Full Test Sets)

Metric	ANN–MNIST	SNN–MNIST	ANN–ECG	SNN–ECG (direct enc.)
Macro-F1	0.968	0.976	0.675	0.706
Latency, optimized C [ms]	13.6	57.1	4.87	8.30
Latency, + coprocessor [ms]	5.75	31.9	2.13	2.66
Worst case, + coprocessor [ms]	10.1	52.0	2.56	3.81
Throughput, + coprocessor [1/s]	174	31.4	469	376
Executed MACs / SOPs	20 167 MAC	106 793 SOP	6 754 MAC	6 639 MAC + 194 SOP
Neuron-state updates	–	3 450	–	670
Model memory [B]	103 024	103 024	9 620	9 620
Neuron state [B]	–	1 104	–	536

MNIST: 2 seeds, 10 000 test images, SNN with $T = 25$. ECG: 3 seeds, 49 273 test beats, SNN with $T = 10$. Coprocessor: MAC unit for the ANN, SNN unit for the SNN. Executed operations count only non-zero inputs and spikes.

TABLE VII: ECG Per-Class Quality on the Test Patients
(49 273 Beats, Mean over 3 Seeds)

		Recall (sensitivity)			Precision		
Network	Macro-F1	N	S	V	S	V	Acc.
ANN (one beat)	0.675	0.968	0.173	0.933	0.297	0.761	93.6 %
SNN (one beat)	0.706	0.971	0.234	0.949	0.387	0.780	94.2 %
Rec. SNN (stream)	0.667	0.971	0.187	0.875	0.336	0.723	93.6 %
GRU (stream)*	0.670	0.964	0.194	0.886	0.338	0.732	93.1 %

Test set: 44 218 N, 1 836 S, 3 219 V beats. *PC floating-point model on the full test set (the GRU was run on the board only on a 1 000-beat subset). One test patient (record 232) contributes 1 381 of the 1 836 S beats, and all networks classify only about 28 % of that patient's beats correctly, so S recall is dominated by this single record.

A. Classification Quality

The classification outputs of the FPGA were identical to the software 8-bit reference model for every test input, seed, and implementation level.

On the testing datasets for MNIST and ECG, both networks exceed the respective macro-F1 targets of 0.95 and 0.65. Specifically for ECG, the macro-F1 difference of 0.031 between the SNN and ANN is about twice the largest seed standard deviation of 0.015.

Table VII shows that this difference is not uniform across classes. Normal beats (N) are recognized with a recall of about 0.96–0.97 by all networks. For ventricular beats (V), recall is 0.93–0.95 for the single-beat models and 0.88–0.89 for the streaming models. Supraventricular beats (S) remain difficult to detect, with a recall of 0.17–0.23 and a precision of 0.30–0.39. The SNN improves both S recall and precision over the ANN, accounting for most of its higher macro-F1. Overall accuracy lies between 93.1% and 94.2% for all networks, only a few points above the 90% obtained by always predicting the N class.

B. Latency and Throughput

Fig. 13 shows the mean latency per input for each implementation level. On MNIST, the optimized C implementation reduces the latency from 136 ms to 13.6 ms for the ANN and from 403 ms to 57.1 ms for the SNN. The optimized C ANN is also 5.6 times faster than the unmodified muRISCV-NN library (76.6 ms). Adding the coprocessor further reduces the latency to 5.75 ms for the ANN and 31.9 ms for the SNN, providing speedups of 2.4 and 1.8 over optimized C.

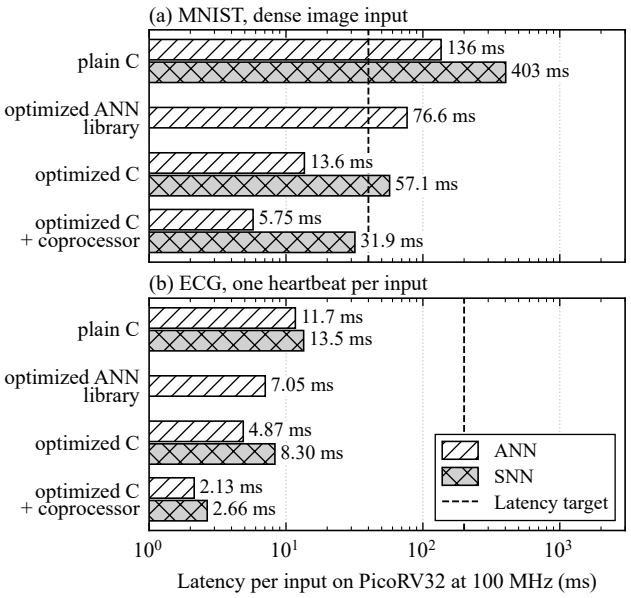


Fig. 13: Measured latency per input on the PicoRV32 for every implementation level: (a) MNIST, (b) ECG with one heartbeat per input. "Optimized library" is the unmodified muRISCV-NN library. The standard deviation over seeds is below 1% and not shown. Dashed lines mark latency targets of Table III.

For ECG, all implementation levels meet the 200 ms target by more than an order of magnitude, including plain C (11.7 ms ANN, 13.5 ms SNN). With the coprocessor, the latencies fall to 2.13 ms for the ANN and 2.66 ms for the SNN. The SNN benefits more from its coprocessor (speedup 3.1 against 2.3 for the ANN), which reduces the latency gap between the two networks from 3.4 ms to 0.5 ms.

The worst-case latencies are listed in Table IX. While all ECG implementations stay below the 200 ms target in the worst case, on MNIST, the SNN with coprocessor exceeds the 40 ms target in its 95th percentile (40.7 ms) and maximum (52.0 ms). Since PicoRV32 processes one input at a time, the throughput is the inverse of the mean latency. Thus, all configurations with coprocessor achieve the throughput target, the lowest being 31.4 images/s for the MNIST SNN against a target of 25 images/s.

Fig. 14 shows how network size and the number of timesteps T trade quality for latency. For the ECG ANN, increasing the hidden layer from 8 to 128 neurons increases the latency without improving the macro-F1, which decreases slightly from about 0.69 to 0.67. The ECG SNN reaches higher macro-F1 values (up to about 0.72) at higher latency, but quality does not increase consistently with T or hidden size. On MNIST, the SNN accuracy increases with T, from about 93 % at $T = 2$ to about 97.6 % at $T = 25$, while the latency increases from below 10 ms to 57.1 ms. The SNN exceeds the ANN accuracy only for $T \geq 8$, where its latency is already higher than that of the ANN.

Fig. 15 combines the ECG quality and latency of all evaluated models. The one-beat SNN with coprocessor reaches

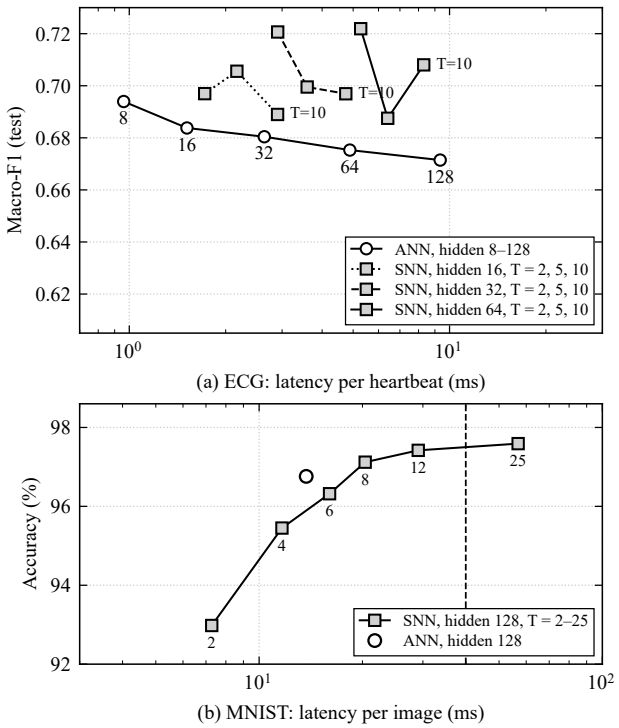


Fig. 14: Quality–latency trade-off for different network sizes and timestep counts T (optimized C on PicoRV32): (a) ECG macro-F1 on the test patients versus latency per heartbeat, (b) MNIST accuracy versus latency per image. Numbers next to the points give the hidden-layer size (ANN) or T (SNN).

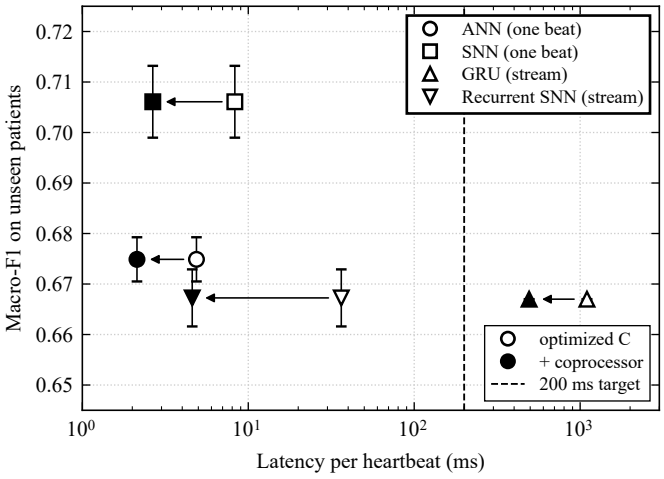


Fig. 15: ECG classification quality versus latency per heartbeat. Open markers: optimized C; filled markers: with coprocessor. All points are full test set measurements except the GRU (1 000-beat subset).

the highest macro-F1 at 2.66 ms, only 0.5 ms slower than the ANN with coprocessor. Among the streaming models, the recurrent SNN with coprocessor meets the 200 ms target, while the GRU stays above it even with its coprocessor.

Fig. 16 compares the two streaming models, which receive the ECG one sample at a time. Per heartbeat, the recurrent SNN executes about 9.6k operations, against 0.10M for GRU-

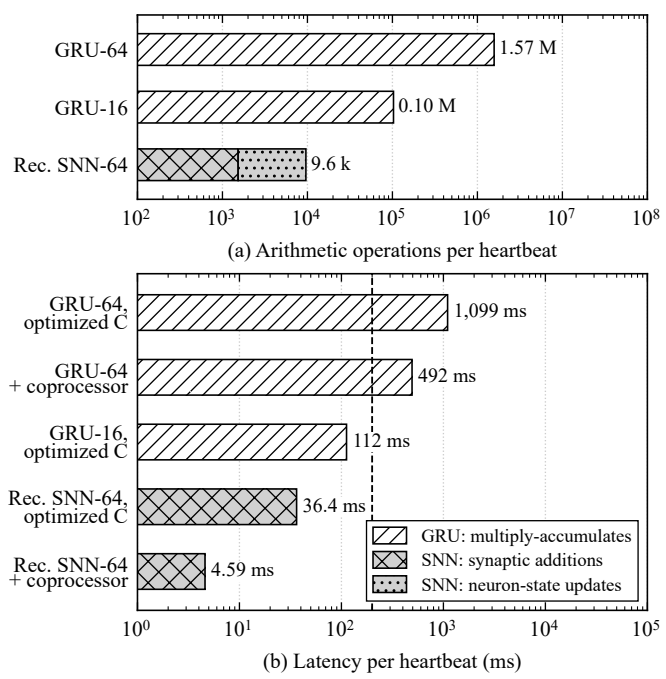


Fig. 16: Streaming ECG inference, one sample per step: (a) arithmetic operations per heartbeat, (b) latency per heartbeat on PicoRV32 for GRU and recurrent SNN (GRU on a 1 000-beat subset).

16 and 1.57M for GRU-64. Its latency is 36.4 ms in optimized C and 4.59 ms with coprocessor, with a maximum of 13.9 ms. GRU-64 therefore misses the 200 ms target even with its coprocessor, and only the smaller GRU 16 meets it, whereas the recurrent SNN stays below it with a large margin of about 14× in the worst case (13.9 ms with coprocessor).

C. Hardware Cost

Table VI lists the executed operations and the memory of each network. Because zero-valued inputs are skipped, the MNIST ANN executes 20 167 of its 101 632 dense MACs. The rate-coded MNIST SNN executes 106 793 SOPs and 3 450 neuron-state updates, about 5.3 times the ANN operation count. For ECG, the SNN executes 6 639 MACs in its first layer and 194 SOPs, plus 670 neuron-state updates, against 6 754 MACs for the ANN. All networks remain within the computational budgets of Table III.

Both networks consume the same model memory for a given dataset. The ECG models require 9 620 B and meet the 32 KB target. The MNIST models require 103 024 B and exceed it by a factor of about 3. The SNNs need an additional 1 104 B (MNIST) and 536 B (ECG) for neuron states.

Table VIII lists the FPGA resources after routing. The MAC unit adds 758 LUTs and 188 flip-flops to the PicoRV32 SoC. The SNN unit adds 889 LUTs, 342 flip-flops and two DSP slices. With both units, the board image uses 4 301 LUTs, about 8 % of the 53 200 LUTs of the Zynq-7020. The BRAM usage is unchanged at 64 tiles, and the 100 MHz clock is met with both coprocessors in place.

TABLE VIII: FPGA Cost of the Custom Units (after Routing)

Design	LUTs	FFs	DSPs	BRAM tiles	WNS (ns)	Unit power (mW, est.)
PicoRV32 SoC	2 631	2 550	6	64	0.63	–
+ MAC unit	3 389	2 738	6	64	0.67	4
+ SNN unit	3 520	2 892	8	64	0.45	13
+ both (board image)	4 301	3 051	8	64	0.80	3 + 10

Increments over the SoC: MAC unit +758 LUTs, +188 FFs. SNN unit +889 LUTs, +342 FFs, +2 DSPs. Positive worst negative slack (WNS): the 100 MHz clock is met in all designs. Power is the Vivado estimate for the unit's hierarchy (default switching activity).

V. Discussion

A. Hypothesis and Hardware Support

The hypothesis expected the ANN to be the better choice for dense inputs (MNIST) and the SNN for sparse, time-varying inputs (ECG), with a small hardware extension making the SNN cost-efficient. The results support the first part, support the second part only for streamed ECG, and support the third part.

For MNIST, the ANN is clearly the better design. The SNN reaches a slightly higher macro-F1 (0.976 against 0.968), but with the coprocessor it is 5.5 times slower (31.9 ms against 5.75 ms), executes 5.3 times more operations, and misses the 40 ms target in the worst case. The rate encoding is the reason: each of the $T = 25$ timesteps repeats the synaptic work for every input spike, so a dense image becomes a larger workload rather than a sparser one. The SNN only exceeds the ANN accuracy for $T \geq 8$ (Fig. 14), where its latency is already above that of the ANN.

For ECG classified one beat at a time, the outcome is mixed. The SNN gives the best classification quality of all models, with a macro-F1 of 0.706 against 0.675 for the ANN, mainly through better detection of S beats. However, it does not deliver the expected latency or operation-count advantage. With the coprocessor it is 1.25 times slower than the ANN (2.66 ms against 2.13 ms), and it performs a similar number of synaptic operations, with additional neuron-state updates. The reason is the direct encoding selected in Section III-D-1: the quantized beat window is fed to the first layer as dense values, so this layer is a regular MAC layer, identical in cost to the ANN. Event-driven computation only takes place in the output layer, which accounts for 194 of the roughly 6 800 operations. The ECG beat window is therefore not sparse from the network's point of view, even though the underlying signal is.

The SNN advantage appears when the ECG is processed as a stream, which is the situation the hypothesis actually describes. Here the recurrent SNN and the GRU reach almost the same macro-F1 (0.667 and 0.670), but the recurrent SNN executes about 9.6k operations per heartbeat against 0.10M for GRU-16 and 1.57M for GRU-64. With its coprocessor it classifies a heartbeat in 4.59 ms, while GRU-64 needs 492 ms. Between consecutive samples few neurons spike, so most synaptic work is skipped. This is where an event-driven model saves work, while a recurrent ANN must evaluate its full gate matrices at every sample. The comparison should be read with the caveat that the GRU was evaluated on the board on a 1 000-beat subset only.

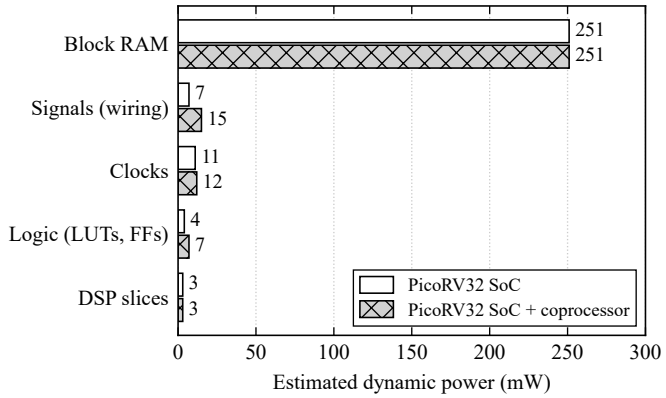


Fig. 17: Estimated dynamic power of the programmable logic of the PicoRV32 SoC with and without the SNN coprocessor (Vivado power report, default switching activity). Not shown and identical in both designs: Arm processing system 1.26 W, device static power 0.15 W.

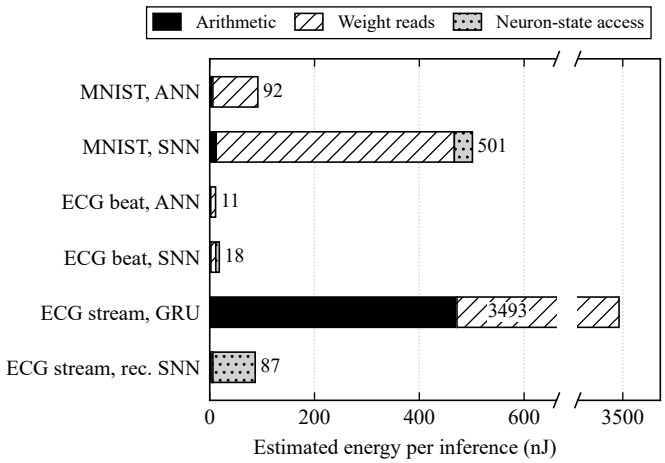


Fig. 18: Estimated energy per inference of each workload on a dedicated 45 nm chip, computed from the measured operation counts and per-operation energies after Horowitz [23]. This is a projection, not a measurement.

The custom hardware support is worth its cost, although for ECG it is no longer needed to meet the 200 ms requirement. The SNN unit speeds up the one-beat ECG SNN by a factor of 3.1 and the recurrent SNN by a factor of 7.9. Since the power on the FPGA is nearly constant, this speedup translates almost directly into lower energy per inference and a longer idle time between beats (Sec. V-B). It costs 889 LUTs, 342 flip-flops and two DSP slices, about 1.7 % of the LUTs of the Zynq-7020, keeps the 100 MHz clock, and adds an estimated 13 mW, about 5 % of the dynamic power of the programmable logic (Fig. 17). Since the inference time drops by a factor of three or more while the power rises by a few percent, the energy per inference also drops. The MAC unit gives the ANN a smaller but similar benefit, a speedup of 2.3–2.4 for 758 LUTs and an estimated 4 mW. In summary, the coprocessor makes the SNN competitive on the single-beat task, but it does not make it faster than an equally accelerated ANN.

Compared with the literature, the macro-F1 values of 0.68–0.71 lie below the 91–99 % reported in Section II-C [11]–[14]. As explained in Section III-D7, our evaluation uses the inter-patient split of de Chazal *et al.* [22], whereas most of those results use intra-patient splits, and several report accuracy rather than macro-F1. Our accuracy of 93.6–94.2 % is in the same range as the 91.4 % of the SNN processor of Mao *et al.* [13], but on this dataset accuracy mainly reflects the majority class.

B. Technology Perspective: From FPGA to Silicon

1) Where the power goes on the FPGA: The power of a CMOS circuit consists of a dynamic, a static and a short-circuit component [25]:

$$P = \tfrac{1}{2}\, C V^2 a f + I_{\mathrm{leak}} V + I_{\mathrm{sc}} V, \tag{7}$$

where C is the switched capacitance, V the supply voltage, a the activity factor and f the clock frequency. Fig. 17 shows that the dynamic power of the programmable logic is dominated by block RAM, which takes 251 mW of the 276 mW of the base SoC, or about 91 %. Logic and DSP slices together account for only 7 mW, and adding the SNN coprocessor raises the total to 288 mW, mostly through additional routing (signals, 7 to 15 mW). The device static power of 0.15 W, the $I_{\mathrm{leak}} V$ term, is about a third of the programmable-logic total. The Arm processing system adds 1.26 W but does not take part in inference.

In terms of Eq. (7), the dominant switched capacitance is that of the memory, not of the arithmetic. This matches the observation in the lecture that a memory access costs about three orders of magnitude more energy than an addition [25]. The SNN coprocessor changes the arithmetic, but the memory term remains the same in both designs. Because the BRAM estimate uses Vivado's default switching activity, it may overestimate the real BRAM power. Nevertheless, the share of memory relative to logic is clear.

2) Why energy follows latency on the FPGA: On this FPGA, the power is essentially constant during inference. The clock reaches all BRAM tiles and logic every cycle, whether a given operation is executed or skipped, and nothing is clock- or power-gated. The energy per inference is therefore approximately $E \approx P \cdot t_{\mathrm{inf}}$. Taking the programmable-logic dynamic and static power of about 0.44 W, one ECG beat costs about 0.93 mJ for the ANN and 1.17 mJ for the SNN with coprocessor, and one MNIST image about 2.5 mJ and 14 mJ. These are estimates under the same assumptions as Fig. 17.

As a consequence, sparsity only saves energy on this platform if it shortens the execution time. A skipped synaptic operation saves the cycles it would have taken, but the memory and clock network keep consuming power during the remaining cycles. Energy savings from sparsity in proportion to the skipped work require hardware in which idle parts can be switched off, for example by clock gating the arithmetic, gating unused memory banks, or power gating the whole core between events. This also explains why the energy ranking on the FPGA is identical to the latency ranking in Fig. 13.

3) Projection to a dedicated chip: To estimate what a dedicated chip would consume, the measured operation counts of Table VI are combined with published per-operation energies at 45 nm [23] (Fig. 18). In this activity-based model, only operations that are actually executed consume energy, which corresponds to ideal gating. One ECG beat then costs about 11 nJ for the ANN and 18 nJ for the SNN, and one MNIST image 92 nJ and 501 nJ respectively. These values are more than four orders of magnitude below the FPGA estimates above, which reflects both the overhead of a soft-core processor on reconfigurable fabric and the absence of gating on the FPGA. This agrees with the flexibility trade-off described in [28]: flexible platforms pay for their generality in efficiency, while a fixed-function design can remove that overhead.

Fig. 18 also shows that weight reads, not arithmetic, dominate the energy of all one-beat and MNIST workloads. In [23], reading a 64-bit word from an 8 KB SRAM costs about 10 pJ, against 0.03 pJ for an 8-bit addition. Replacing multiplications by additions, which is the usual argument for SNNs, therefore saves little as long as every synaptic operation still fetches its weight. The SNN costs more than the ANN on both one-beat tasks because it executes as many or more operations and additionally reads and writes neuron states. The recurrent SNN on the ECG stream is the exception: at 87 nJ per heartbeat, it needs about 40 times less energy than the GRU (3 493 nJ), because the GRU's cost is dominated by its large number of multiplications. The projection excludes control logic, instruction fetch and leakage, so the values describe lower bounds, while the ratios between workloads are the more reliable result.

4) Neuromorphic chips and in-memory computing: Dedicated neuromorphic processors address the weight-read problem by placing synaptic memory next to the neurons that use it, so that a spike only triggers local memory accesses. IBM TrueNorth, fabricated in a 28 nm CMOS process, distributes its synaptic SRAM over 4 096 cores [29]. Intel Loihi, in a 14 nm FinFET process, follows the same principle and adds on-chip learning [30]. For ECG specifically, the DYNAP-SE processor, which uses analog subthreshold neuron circuits in a 180 nm process, has been used for real-time anomaly detection at very low power [15], and Mao *et al.* present a digital SNN processor for ECG detection [13]. These chips show that the technology node is secondary: an event-driven architecture with local memory is what makes spiking efficient.

In-memory computing goes one step further by performing the multiply-accumulate inside the memory array itself. In a crossbar of resistive or phase-change memory cells, each cell stores a weight as a conductance. Input voltages on the rows then produce currents that are summed on the columns, implementing many multiply-accumulates in a single analog step [31]. This removes the weight read altogether, but it benefits dense ANN layers as much as spiking ones. The 2D materials mentioned in Section II-F aim at the same goal by lowering the switching energy of synaptic devices [17].

5) Requirements of a wearable ECG device: At a resting heart rate of about 70 beats/min and 2.66 ms per beat, the processor would be active for only about 0.3 % of the time, and still for only about 1 % at the 240 beats/min of the operating envelope in Section II-D. With the projected 18 nJ per beat, the inference energy amounts to about 20 nW on average, four orders of magnitude below the 670 μW of the ADS1292R front end [24]. In a wearable device, the inference arithmetic is therefore not the limiting factor. Instead, idle power consumption of the analog front end and the leakage of logic are. The Vivado estimates therefore cannot predict the battery life of such a device: they describe an FPGA that draws about 0.44 W continuously, whereas a wearable monitor would be dominated by the sub-milliwatt front end and by leakage during the idle time between beats.

Such a device would therefore need (i) power gating of the processor between beats, woken by the R-peak detector; (ii) on-chip SRAM that retains the weights at a reduced voltage, or non-volatile memory that needs no retention power [26] implemented in (iii) a low-leakage process. Leakage became a major design concern after the end of Dennard scaling, and techniques such as high-κ metal gates and FinFETs were introduced partly to reduce it [25], [26]. For a duty-cycled design of this size, a mature low-power node such as the 65 nm CMOS proposed in Section II-F is sufficient, since the small active area makes leakage, not density, the main concern. Finally, the design should keep the microarchitecture simple: as shown for Intel processors, added complexity increases the energy per instruction more than it improves performance [27], which favors a small in-order core with a dedicated neuron unit, as used here.

The inference energy estimates do not represent the total energy consumption of the proposed device. A complete device power requirements would also need to account for continuous ECG acquisition, preprocessing, communication, memory retention, and power-conversion losses.

C. Limitations

Several limitations affect how far these conclusions can be generalized. First, all reported power and energy values are estimates. The FPGA estimates use Vivado power reports with default switching activity, while the 45 nm projections combine operation counts with published per-operation energies. The latter exclude control logic, instruction fetch, and leakage. No power was measured at the board supply. Second, all results were obtained on a single in-order soft-core processor at 100 MHz. A parallel or pipelined accelerator could change the latency ratios between the networks, since dense MAC layers parallelize more easily than event-driven updates.

Third, the number of seeds is small (two for MNIST, three for ECG). The seed standard deviation of at most 0.015 suggests that the ECG macro-F1 difference of 0.031 is real, but more seeds would be needed to confirm it. Fourth, the ECG test set is strongly imbalanced, and a single record (232) contributes 75 % of all S beats, so the S-class results largely reflect one patient. Although all one-beat models meet the 0.65 macro-F1 target, an S recall around 0.2 would not be acceptable for clinical use. Fifth, the GRU was run on the board only on a 1 000-beat subset. Finally, the MNIST models exceed the 32 KB memory target by a factor of three, so

the MNIST comparison does not fully represent a memory-constrained edge device.

VI. Conclusion

This project investigated whether spiking neural networks (SNNs) provide advantages over artificial neural networks (ANNs) for hardware-constrained edge inference, and whether dedicated hardware support can improve their efficiency. Both architectures were evaluated on a PicoRV32 RISC-V soft-core at 100 MHz on the PYNQ-Z2 board, using MNIST as a reference benchmark and ECG arrhythmia classification as the main application.

For MNIST, the ANN proved to be faster. Although the SNN achieved a slightly higher macro-F1 score (0.976 vs. 0.968), it was 5.5 times slower with hardware acceleration. The SNN also executed 5.3 times more operations, demonstrating that rate encoding introduces substantial computational overhead for dense image inputs.

For single-beat ECG classification, the SNN achieved a higher macro-F1 score of 0.706 compared with 0.675 for the ANN, mainly through improved detection of supraventricular beats, but remained 1.25 times slower despite hardware acceleration. All single-beat configurations meet the 200 ms latency target by more than an order of magnitude, so for this task the network choice is decided by classification quality and energy rather than speed. A clearer advantage appears when processing ECG as a continuous stream. The recurrent SNN classifies a heartbeat in 4.59 ms, whereas GRU-64 needs 492 ms and misses the target even with hardware acceleration.

These results confirm the hypothesis for dense inputs, where the ANN is the better choice, and for streamed ECG, where the SNN is more suited. For single-beat ECG, the hypothesis is only partly confirmed. Direct encoding feeds the entire beat to the first layer as dense values, so a sparse signal in time is not sparse for the network. The custom coprocessor confirms the third part of the hypothesis, as it speeds up the SNN three- to eightfold for a small amount of additional FPGA logic.

Overall, SNNs are not universally more efficient than ANNs. Their advantages depend on input representation, spike activity, and hardware implementation. Estimated power consumption was dominated by memory accesses rather than arithmetic operations, indicating that reducing computational operations alone does not guarantee proportional energy savings.

Future work should quantify the trade-offs of a fully event-driven single-beat SNN, use delta encoding as in the recurrent SNN instead of direct encoding, apply the stronger down-sampling suggested by the down-sampling study, measure consumed power instead of estimating it, and add clock and power gating, before validating the system with continuous ECG acquisition. To implement either network in silicon, the energy of weight reads should be reduced, for example by in-memory computing, which would benefit both network types and allow the sparsity of the SNN to translate into proportional energy savings.

Appendix A

TABLE IX: Mean and Worst-Case Latency on the Board (PicoRV32, 100 MHz)

Workload	Implementation	Mean (ms)	p95 (ms)	Max (ms)	Target met
MNIST	ANN, optimized C	13.6	18.7	25.6	yes
	ANN + coprocessor	5.75	7.60	10.1	yes
	SNN, optimized C	57.1	66.8	79.1	no
	SNN + coprocessor	31.9	40.7	52.0	mean only
ECG beat	ANN, optimized C	4.87	5.70	5.94	yes
	ANN + coprocessor	2.13	2.43	2.56	yes
	SNN, optimized C	8.30	9.52	10.2	yes
	SNN + coprocessor	2.66	3.31	3.81	yes
ECG stream	GRU-64, optimized C[a]	1 099	1 103	1 106	no
	GRU-64 + coprocessor[a]	492	494	495	no
	Rec. SNN, optimized C[a]	36.4	43.3	57.2	yes
	Rec. SNN + coprocessor	4.59	6.43	13.9	yes

Targets (Table III): MNIST 40 ms, ECG 200 ms. Full test sets (10 000 images, 49 273 beats) except [a]: 1 000-beat subset. Coprocessor: MAC unit for the ANN/GRU, SNN unit for the SNN (ECG SNN: first layer on the MAC unit).

List of Abbreviations

Abbreviation	Meaning
ADC	Analog-to-Digital Converter
AFib	Atrial Fibrillation
ANN	Artificial Neural Network
ASIC	Application-Specific Integrated Circuit
BRAM	Block Random Access Memory
CMOS	Complementary Metal-Oxide-Semiconductor
CNN	Convolutional Neural Networks
ECG	Electrocardiogram
FPGA	Field-Programmable Gate Array
GRU	Gated Recurrent Unit
IC	Integrated Circuit
LUT	Look-Up Table
MAC	Multiply-Accumulate
MIT-BIH	Massachusetts Institute of Technology – Beth Israel Hospital
MNIST	Modified National Institute of Standards and Technology
MtM	More than Moore
RISC-V	Reduced Instruction Set Computer, Version Five
SNN	Spiking Neural Network
SoC	System-on-Chip
SPI	Serial Peripheral Interface
DSP	Digital Signal Processing
DUV	Deep Ultraviolet
EUV	Extreme Ultraviolet
LIF	Leaky Integrate-and-Fire
PCPI	Pico Co-Processor Interface
SOP	Synaptic Operation

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